



جامعة نيويورك أبوظبي
NYU ABU DHABI

Robotics Lab Report

Course: ENGR-UH 4330| Fall 2025

Experiment: SIR Model – Flu Epidemic Simulation in Simulink

Date: 30/ 09 / 2025

Group Members: Demarce Williams

Part 1: Rotation Matrices and Homogeneous Transformations

Initial Rectangle: Geometry Setup

Visualization of the base rectangle geometry before any transformations are applied.

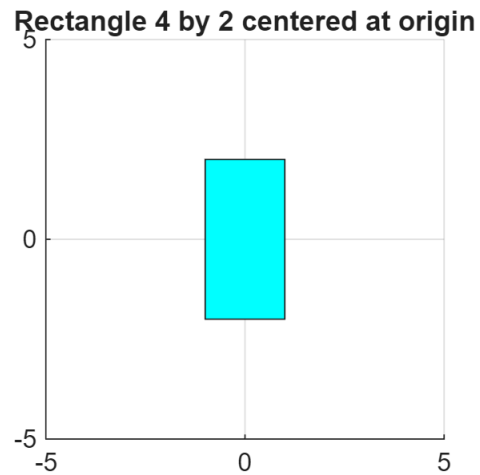


Figure 1. Initial 4x2 Rectangle Centered at Origin (0,0)

Rotation Control

Interactive slider controls the angle θ to rotate the rectangle around the z-axis using a custom-defined rotation matrix.

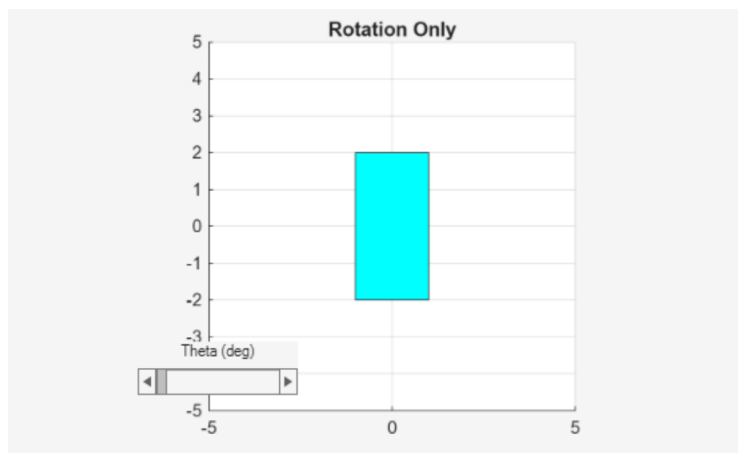


Figure 2: Rotation of Rectangle About Z-Axis Using Theta Slider

Rotated Rectangle Using Custom Matrix

Demonstrates the effect of manual rotational transformation on the rectangle.

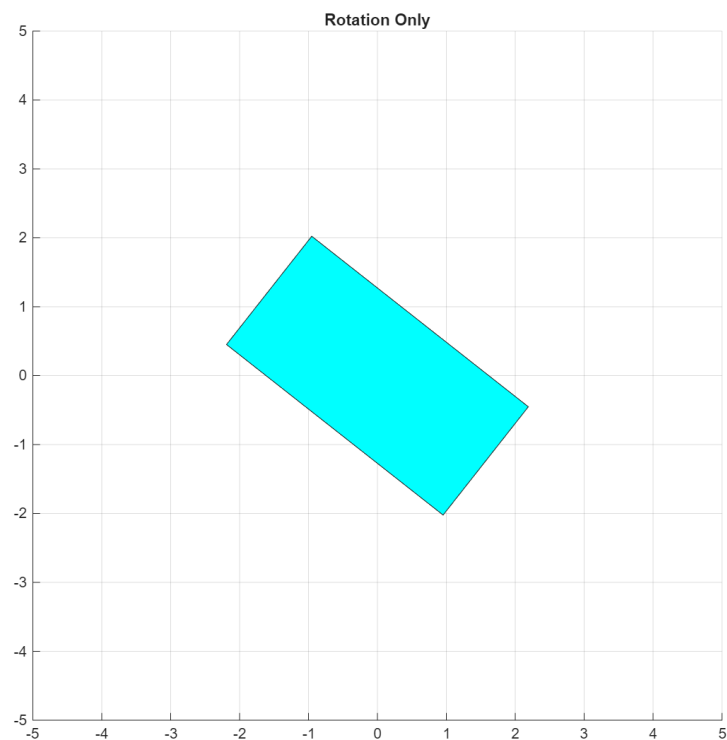


Figure 3: Rectangle Rotated Around Z-Axis Using Manual Rotation Matrix

Translation Controls

Interactive sliders for translating the rectangle in x and y directions.

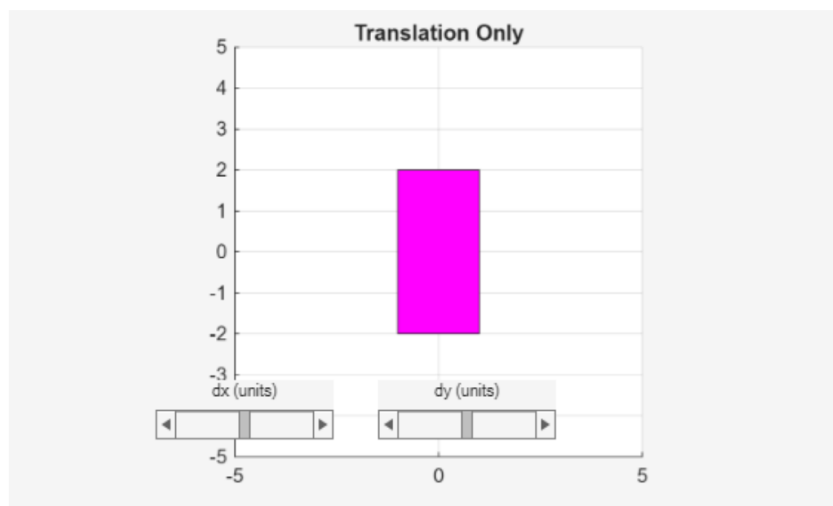


Figure 4: Translation of Rectangle Along X and Y Axes Using Sliders (dx, dy)

Translation Along X-Axis

Rectangle shifted horizontally using the value from dx slider.

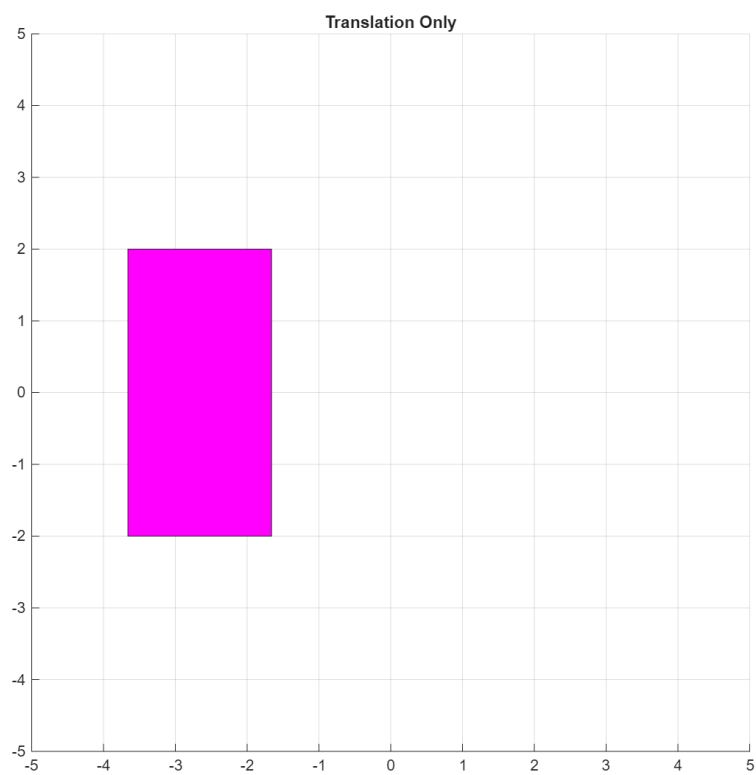


Figure 5: Rectangle Translated Along X-Axis by dx Units

Translation Along Y-Axis

Rectangle shifted vertically using the value from dy slider.

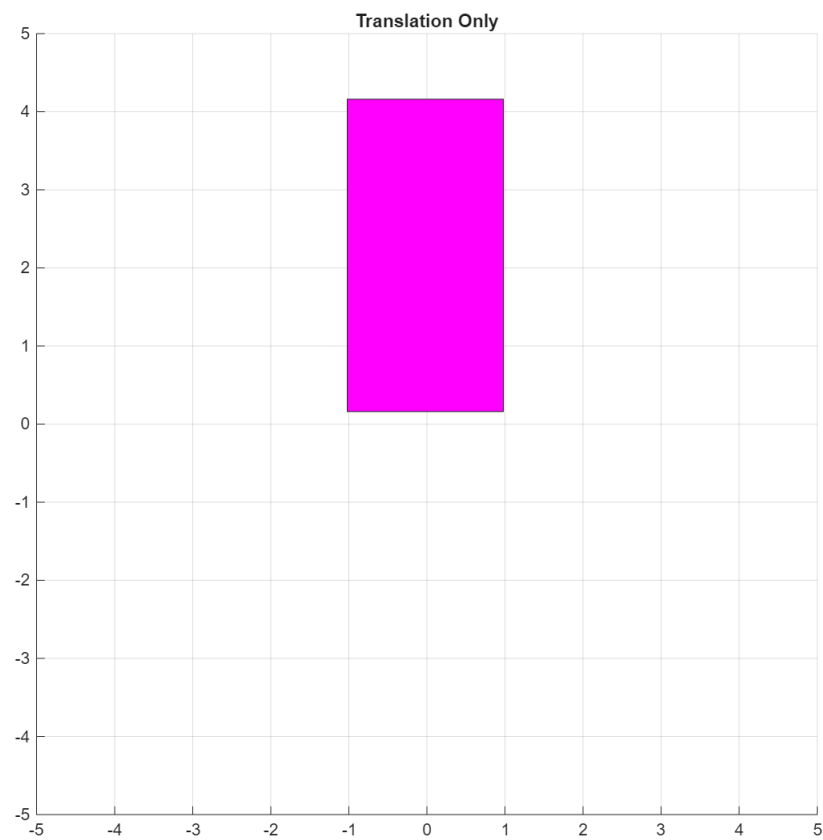


Figure 6: Rectangle Translated Along Y-Axis by dy Units

Combined Transformation Controls

Interface for simultaneous rotation and translation of the rectangle using homogeneous transformation matrix.

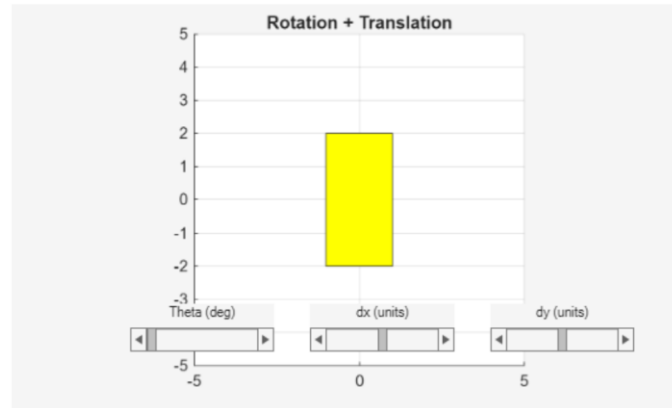


Figure 7: Combined Rotation and Translation Using Sliders for θ , dx , and dy

Rectangle After Combined Transformation

Visualization of the final transformed rectangle after applying both rotation and translation.

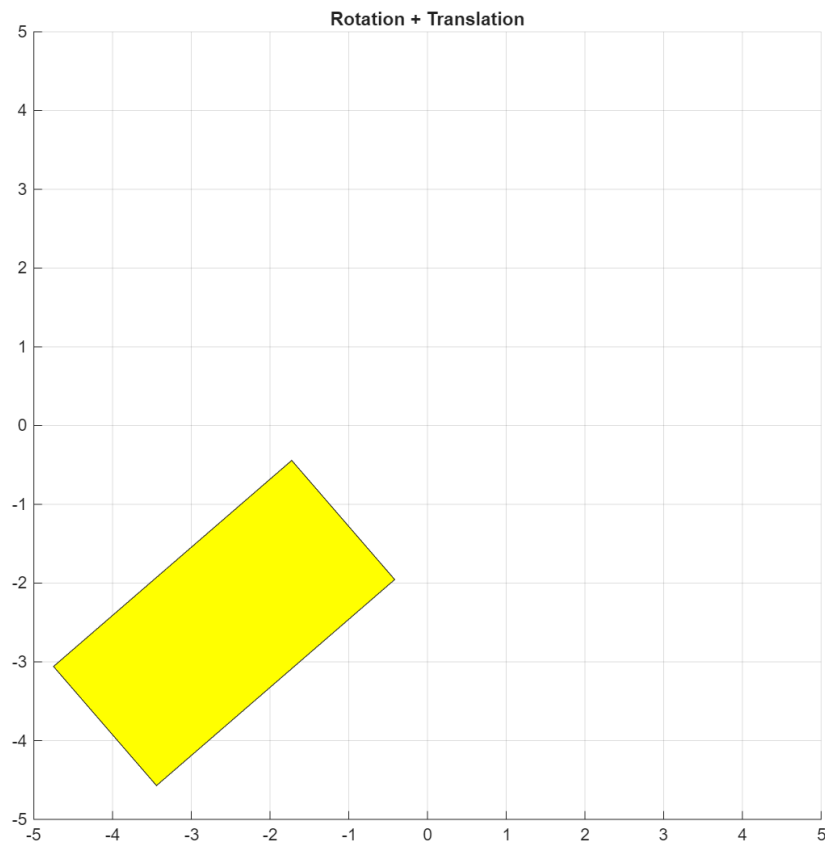


Figure 8: Transformed Rectangle Using Combined Rotation and Translation Matrix

Using MATLAB Built-in Rotation Functions

Comparison between manual and built-in MATLAB `rotz()` rotational transformations.

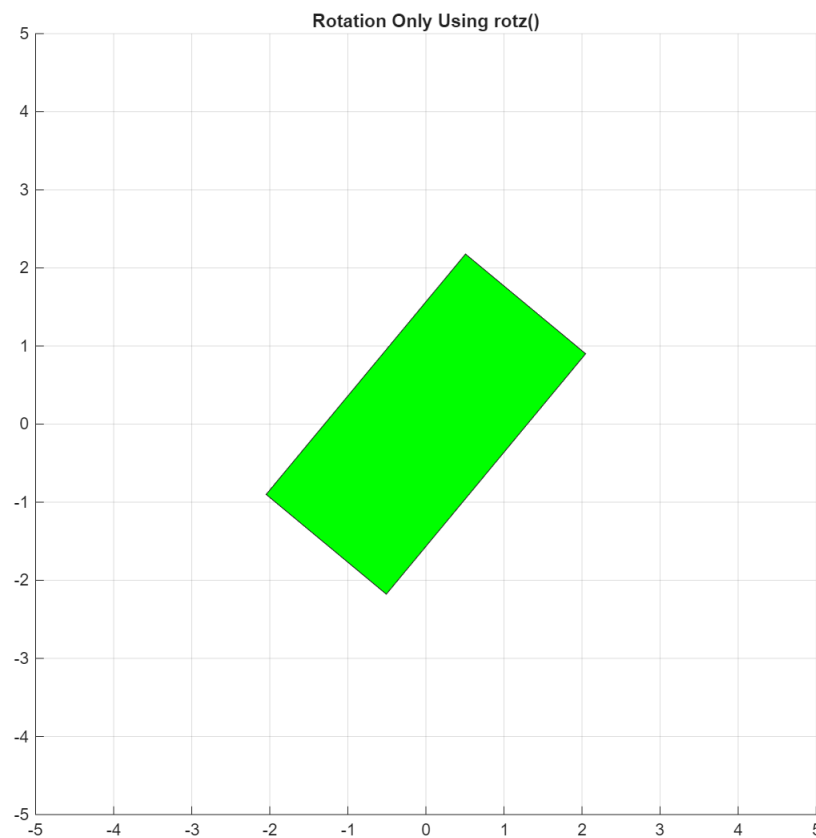
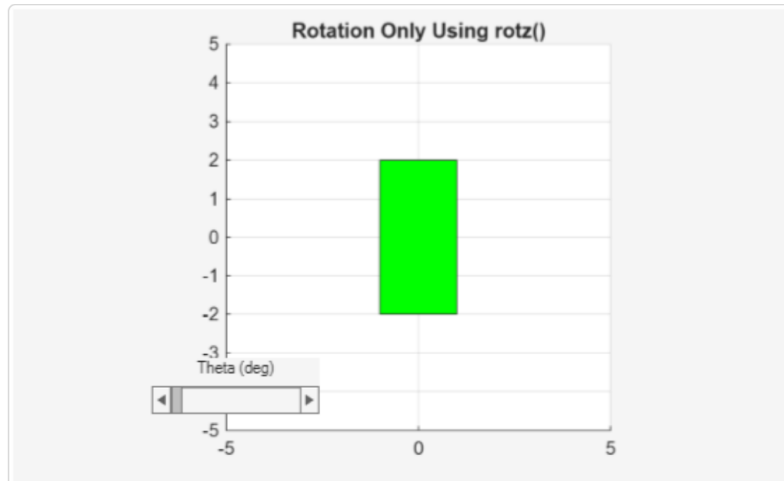


Figure 9: Rotation of Rectangle Using MATLAB's Built-in `rotz()` Function

Part 2: Forward Kinematics of a 2-Link Planar Manipulator

Manipulator Visualization and Joint Control

Visual representation of a planar RR manipulator with sliders to control each joint's angle.

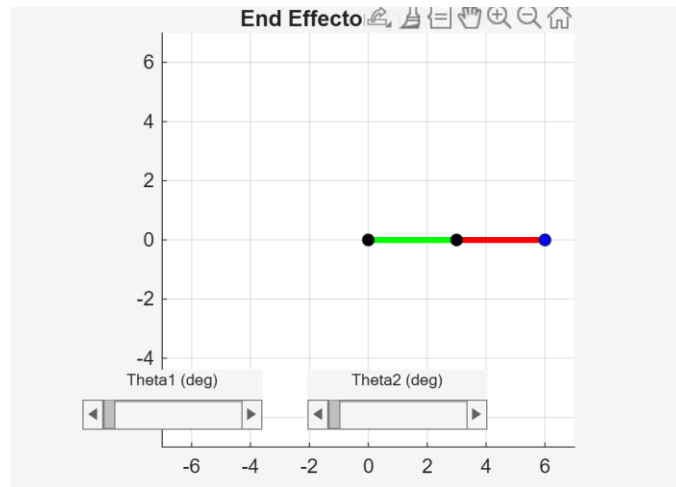


Figure 10: 2-Link Planar Manipulator with Interactive Joint Angle Sliders (θ_1 , θ_2)

Forward Kinematics and End-Effector Position

Live calculation and display of the end-effector position as the joint angles are manipulated

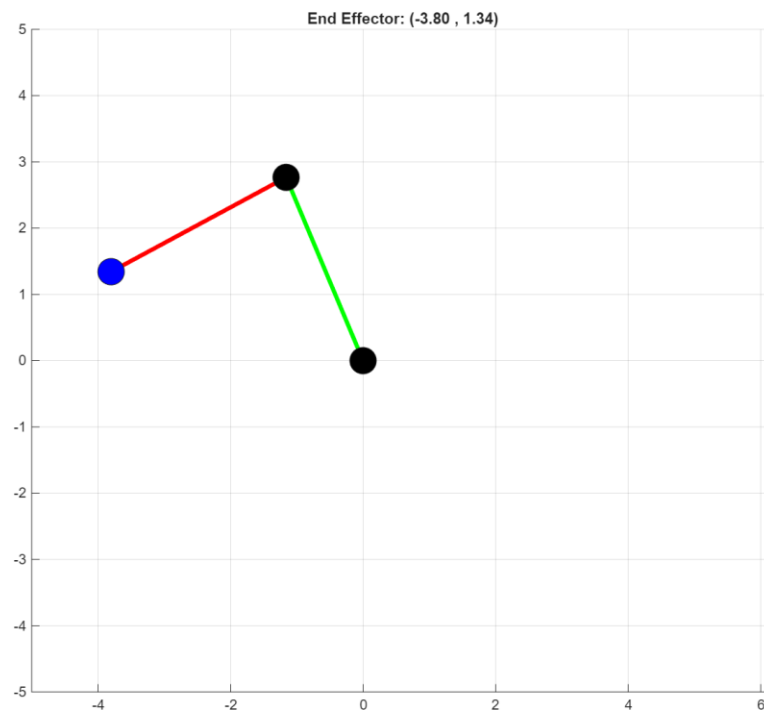


Figure 11: End-Effector Position Updated Dynamically Based on Joint Angles

AVI Animation Output

Exported AVI animation showing continuous movement of the end-effector as joint angles are varied across specified ranges.

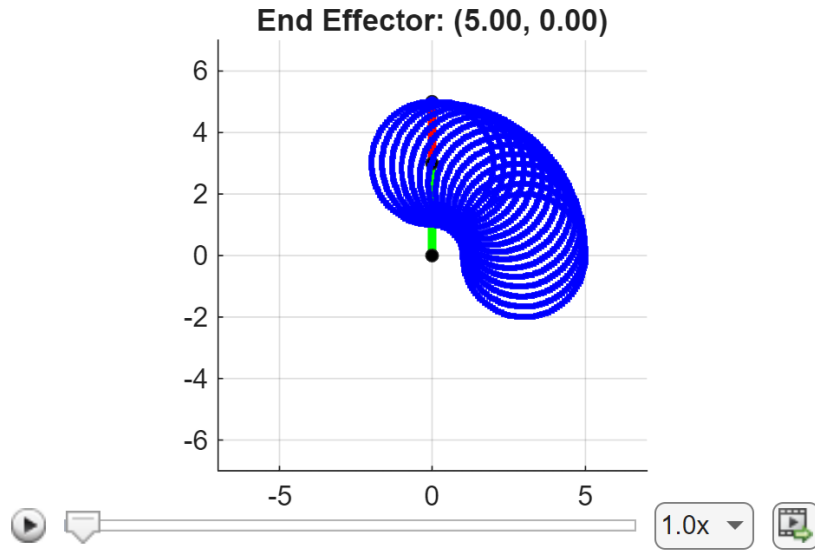


Figure 12: Animation of 2-Link Manipulator Motion: $\theta_1 \in [0, 2\pi], \theta_2 \in [0, 2\pi]$